

Table 1: Published-reference comparison on the 64-trajectory KdV 40s FNO(AR) benchmark. Published values are transcribed from Brandstetter et al. (2022), Table 3, and were not rerun locally. Local rows use five paired training seeds of an independent implementation aligned to the released 20-to-20-step FNO code. LOCO applies Galilean consistency only (g_3 , $\lambda_{\text{orb}} = 0.05$, $|\epsilon| \leq 0.2$); published LPSDA applies $g_1 g_2 g_3 g_4$. Published uncertainty is the reported bootstrap ± 2 SD interval, whereas local uncertainty is sample standard deviation across training seeds. Orbit-OD and equivariance-defect values are reported only where all five comparable evaluations exist. Dashes mean unavailable, not zero. The published and local blocks are protocol-aligned references, not a controlled method-only comparison.

Evidence	Method	Seeds	Params (M)	ID NMSE	Orbit-OD NMSE	Eq. defect	Train h	Latency ms/sample
Published ref.	FNO (AR), no augmentation	–	–	0.1248 ± 0.0108	–	–	–	–
Published ref.	FNO (AR) + LPSDA, $g_1 g_2 g_3 g_4$	–	–	0.0606 ± 0.0040	–	–	–	–
Ours (independent)	FNO (AR), local calibration	5	10.86	0.1205 ± 0.0035	–	–	4.29 ± 0.36	11.90 ± 1.11
Ours (independent)	FNO (AR) + LOCO, g_3 , $\lambda = 0.05$	5	10.86	0.1199 ± 0.0027	0.5341 ± 0.0206	0.5724 ± 0.0160	7.44 ± 0.48	10.15 ± 0.23